

Lab 07 Report

Five-Transistor OTA Design

1. Part 1: OTA Design

Design Specifications

We want to design a 5T OTA using the g_m/I_D methodology to meet the following specs.

Parameter	Value
Supply voltage	1.8V
Load	5pF
Open loop DC voltage gain	$\geq 34\text{dB}$
CMRR @ DC	$\geq 74\text{dB}$
Phase margin (PM)	$\geq 70^\circ$
Current seed (reference current)	$10\mu\text{A}$
CM input range - low	$\leq 1.2\text{V}$
CM input range - high	$\geq 1.5\text{V}$
GBW	$\geq 5\text{MHz}$
OTA current consumption	Minimize
Area	Minimize

Input Pair Selection

Q: From the CMIR spec you need to identify the input pair type: NMOS or PMOS?

We need an NMOS input pair since the input range closer to VDD makes the NMOS in saturation while the PMOS is either in triode or cutoff.

Detailed Design Procedure & Hand Analysis

Explain why you chose the architecture that you implemented and detail your sizing calculations (GBW, tail current, g_m/I_D selections, etc.).

First of all, We will use Single stage OTA, 34dB ≈ 50.11 V/V which is achievable in our GF180 technology. We will use $\frac{g_m}{g_{ds}}$ knob at 100. Also, no compensation network is required to guarantee stability since there is only one H.I.N., so the phase margin is high enough.

Ignoring parasitic capacitances in g_m calculation.

$$g_m = \text{GBW} \cdot 2 \cdot \pi \cdot C_L = 157 \mu\text{S}$$

For the input pair, we need high gmoverid to provide gain. But we don't have specs on the flowing current in the 5T-OTA.

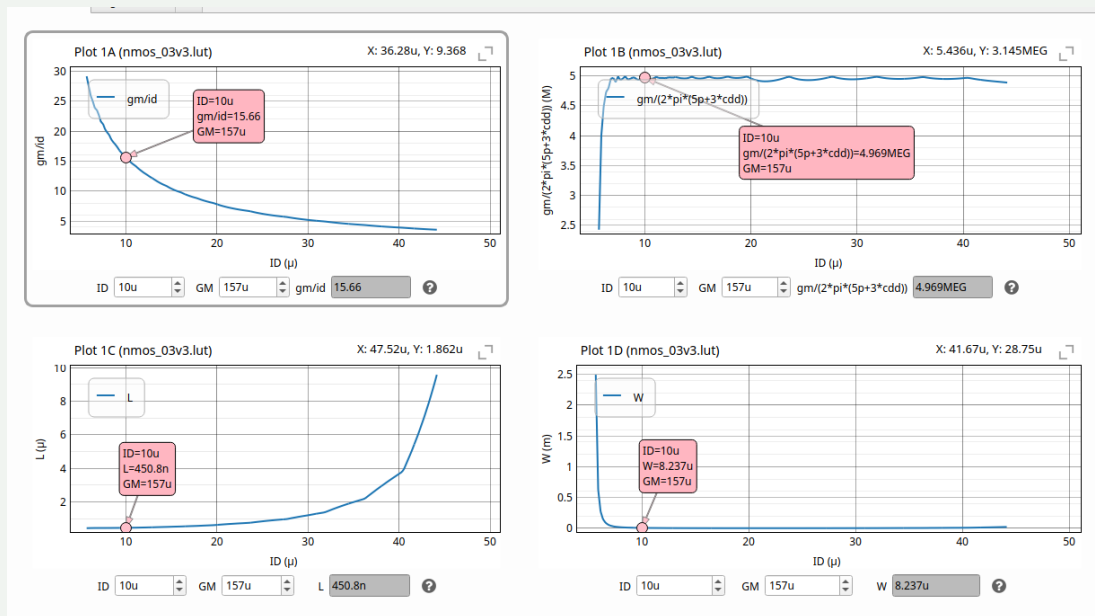


Figure 1: Sizing the input pair, optimizing area, and constant GBW

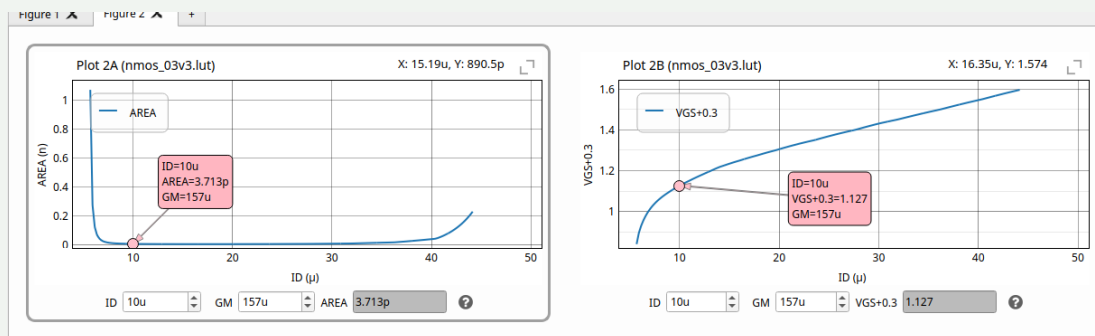


Figure 2: Sizing with area and CMIR considered

For the active load, we need low gmoverid since they don't contribute to gain with g_m , and to save area as well.

Import: Plot 1A OK ?

▼ LUT Settings

LUT: pmos_03v3 ?

Corner: TT ? ☐ All ?

Temp (°C): 27.0 ? ☐ All ?

Frequency: 1 ?

ID: 10u ?

gm/ID: 10 ?

gm/gds: 100 ?

VDS: 0.5 ?

VSB: 0 ?

Stack: 1 ?

Results:

	Name	TT-27.0
1	ID	10u
2	IG	N/A
3	L	450n
4	W	7.52u

Y-Expr: gm/ID*IT ?

Figure 3: Sizing the active load

As for the current mirror, We need low gmoverid for minmizing the error in current copying due to small perturbation (mismatch) between the two transistors.

$$\frac{\Delta I_D}{I_D} = \frac{g_m}{I_D} \cdot \Delta V_{GS}$$

▼ LUT Settings

LUT: nmos_03v3 ?

Corner: TT ? ☐ All ?

Temp (°C): 27.0 ? ☐ All ?

Frequency: 1 ?

ID: 10u ?

gm/ID: 10 ?

L: 1u ?

VDS: 0.3 ?

VSB: 0 ?

Stack: 1 ?

Results:

	Name	TT-27.0
3	L	1u
4	W	4.64u
5	VDB	300m
6	VGB	836.3m

Figure 4: Tail Current source sizing

[04-08-2026 16:59:58] - Info: [Sizing Assistant] The resultant point 'sqrt(1.5)*idms/id*100 = 2.936'
 [04-08-2026 19:00:23] - Info: [Sizing Assistant] The resultant point 'LAMBDA = 99.38m'

Figure 5: Acheived lambda and mismatch from sizing

Another sizing strategy instead of gmoverid I used was to optimise mismatch and lambda, But it resulted in high W, but will provide higher CMRR, differential gain, and lower common mode gain, and a slightly lower phase margin.and GBW.

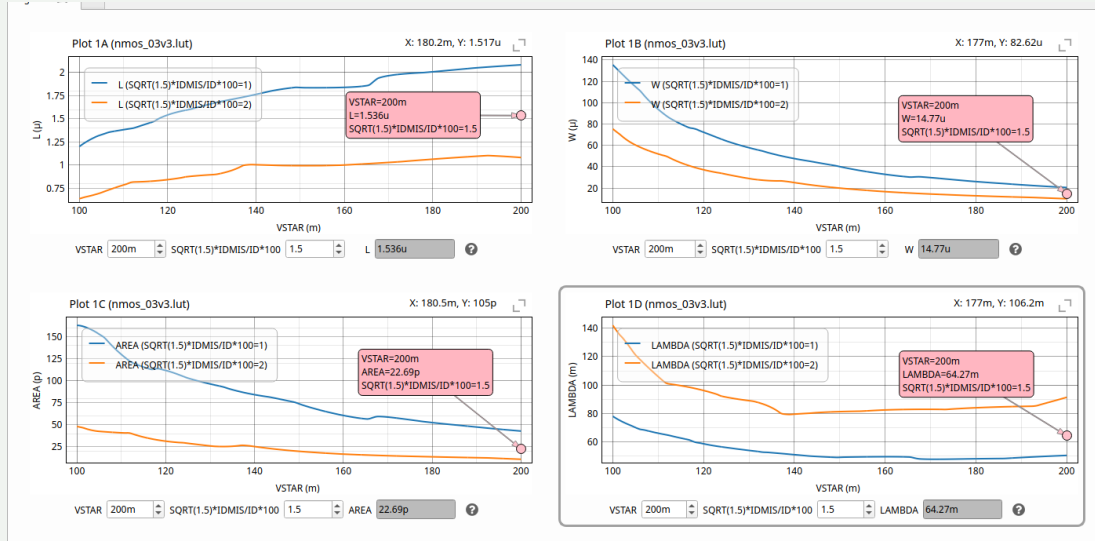


Figure 6: Sizing the current mirror at $V_{DS}=300\text{mV}$ and $I_D=20\mu\text{A}$

Table 1: Final Transistor Sizing and Operating Points

Device	W (μm)	L (μm)	g_m (μS)	I_D (μA)	g_m/I_D (S/A)	V_{dsat} (mV)	V_{ov} (mV)	V^* (mV)
Input Pair (M1, M2)	8.24	0.45	158	10.12	15.61	104	30.7	128.1
Active Load (M5, M6)	7.52	0.45	99.23	10	9.923	174.7	156.4	201.6
Tail Current (M3)	9.32	1	197.5	20	9.876	172.5	148.9	202.5

2. Part 2: Open-Loop OTA Simulation

2.1 1. Schematic and DC Operating Point

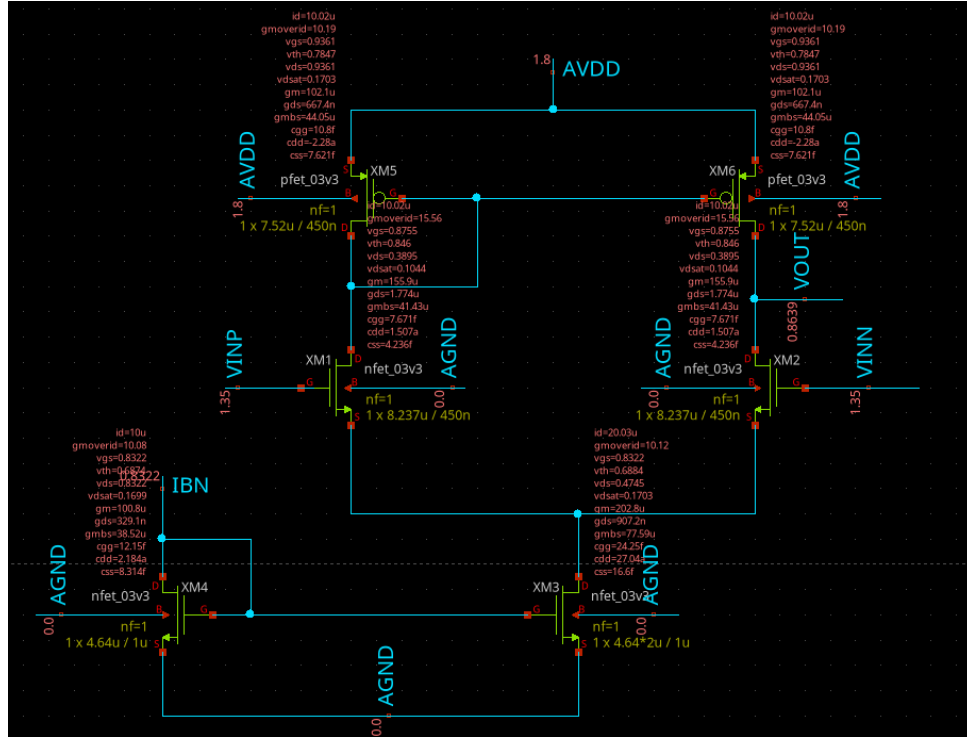


Figure 7: DC Operating points

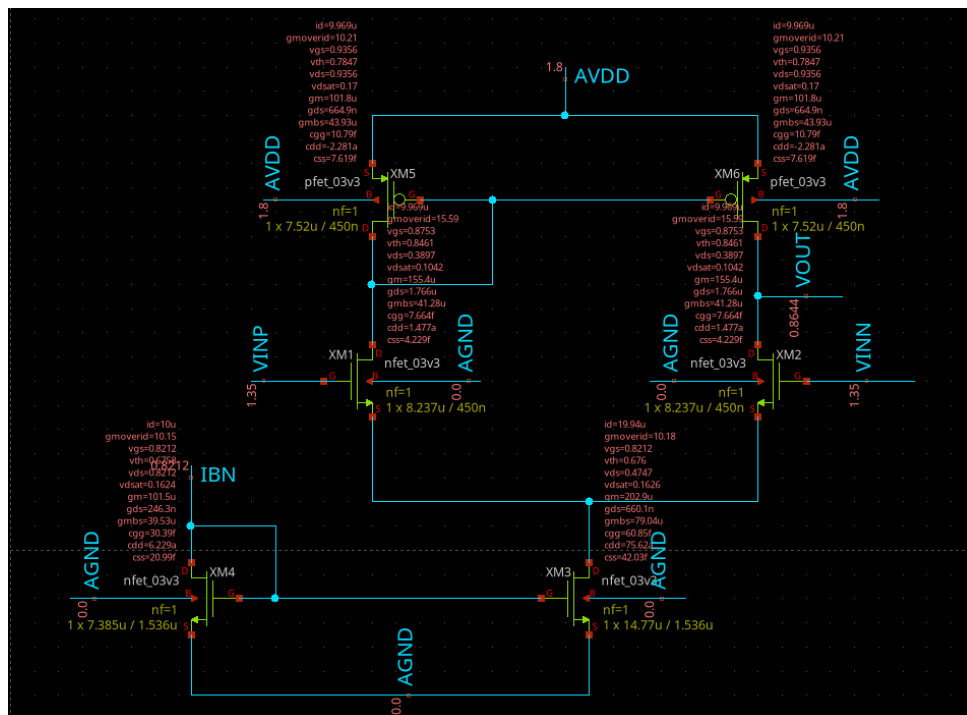


Figure 8: My other sizing for CM

DC Operating Point Analysis

Q: Is the current (and g_m) in the input pair exactly equal? What is the DC voltage at VOUT? Why?

Yes, They are equal as $V_{GS1} = V_{GS2}$ and V_{out} should be equal to V_{mirror} to satisfy $I_{D1} = I_{D2} = I_{D5} = I_{D6}$. Consequently, $V_{DS1} = V_{DS2}$. So they are exactly equal.

$$V_{out} = V_{DD} - V_{SG5,6} = 1.8 - 0.9361 = 0.8639V$$

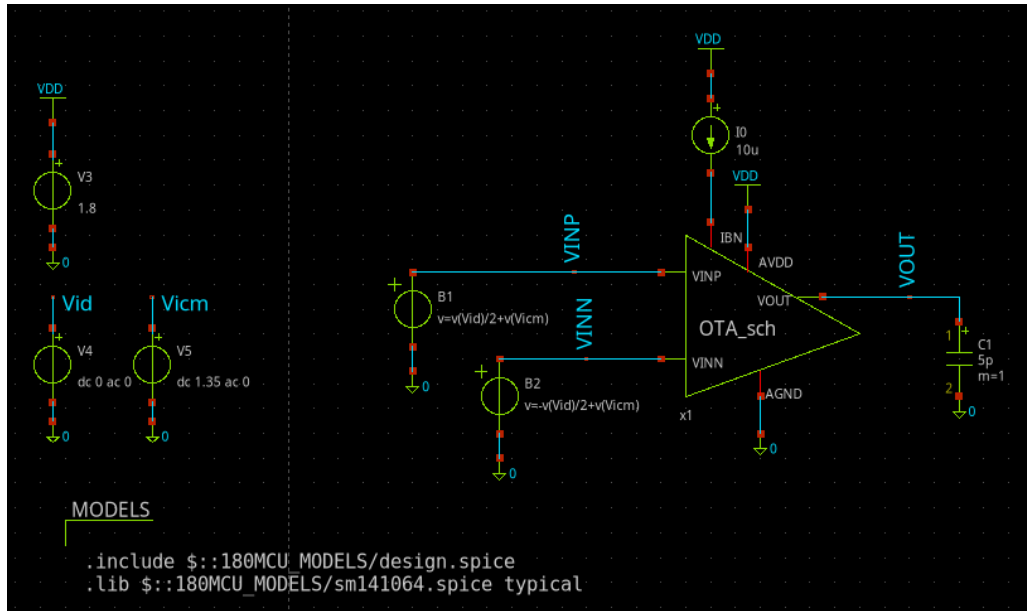


Figure 9: Testbench

2.2 2. Differential Small Signal Characteristics

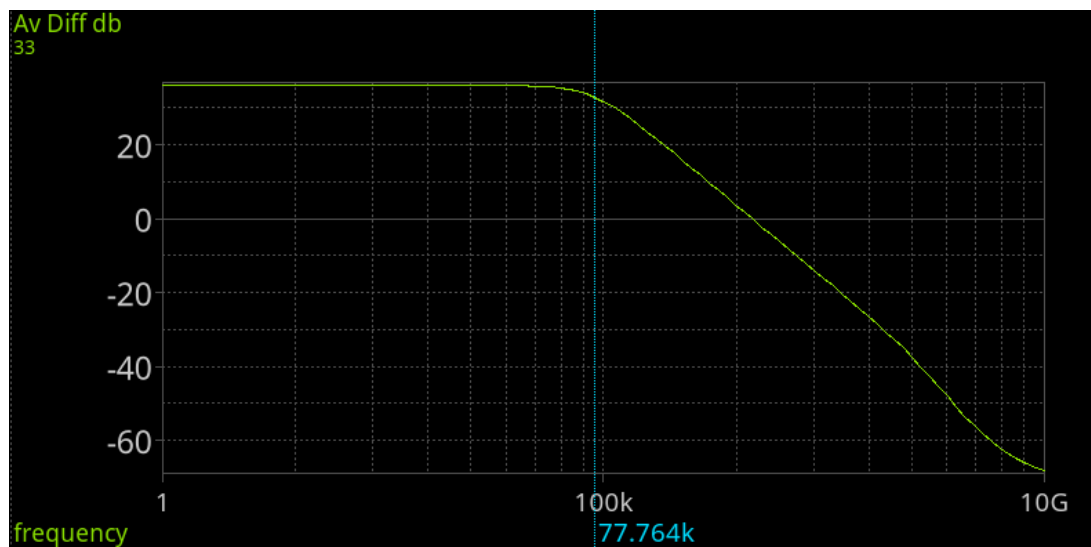


Figure 10: Differential Small Signal Gain Bode Plot

Differential Gain: Hand Calc vs Simulation

Parameter	Simulation	Hand Analysis
DC Gain (A_{vd} dB)	36.077	36.104
GBW (MHz)	4.8967	4.9583
Phase Margin (PM)	90.523	-

2.3 3. Common-Mode Small Signal Characteristics

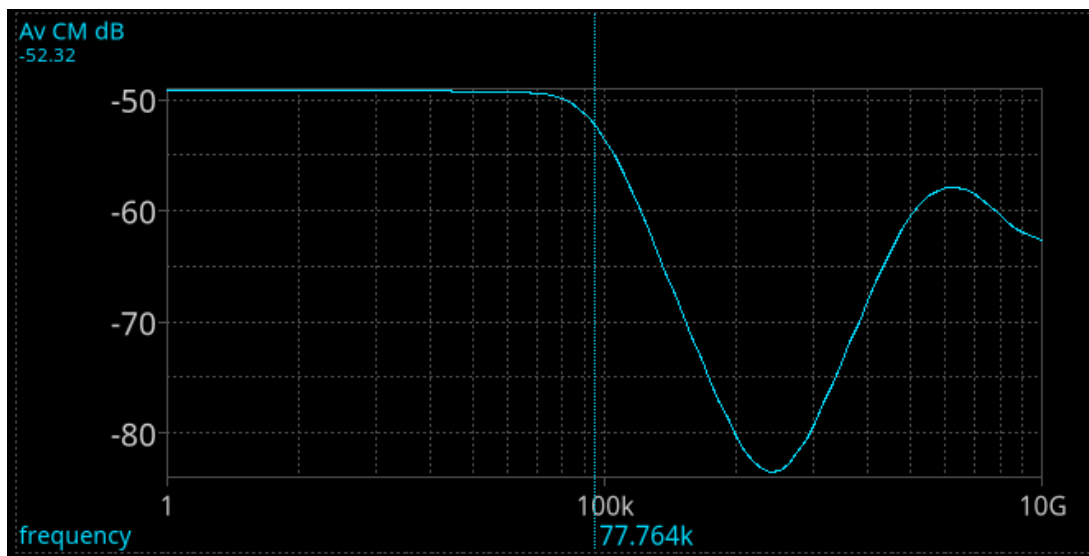


Figure 11: Common Mode Small Signal Gain Bode Plot

Common-Mode Gain: Hand Calc vs Simulation

Parameter	Simulation	Hand Analysis
CM DC Gain (A_{vcm} dB)	-49.244	-47.047

```

=== Bias Points ===
M1: gm/id = 15.5888 S/A, Vstar = 0.128297 V
M2: gm/id = 15.5888 S/A, Vstar = 0.128297 V
M3: gm/id = 10.1745 S/A, Vstar = 0.196571 V
M4: gm/id = 10.1471 S/A, Vstar = 0.1971 V
M5: gm/id = 10.2096 S/A, Vstar = 0.196895 V
M6: gm/id = 10.2096 S/A, Vstar = 0.196895 V
Doing analysis at TEMP = 27.000000 and TNOM = 27.000000

Using SPARSE 1,3 as Direct Linear Solver
No. of Data Rows : 101
Doing analysis at TEMP = 27.000000 and TNOM = 27.000000

Using SPARSE 1,3 as Direct Linear Solver
Reference value : 1.000000e+00
No. of Data Rows : 101
av_diff_db = 3.608475e+01
av_diff_mag = 6.371457e+01
bw_diff = 7.660444e+04
ugf = 4.898203e+06
p_ugf = -8.947759e+01
av_cm_db = -5.197626e+01
av_cm_mag = 2.518762e-03
bw_cm = 7.664327e+04
=== AC Results ===
Av Diff: 36.0847 dB, BW: 76604.4 Hz
UGF: 4.8982E+06 Hz
GBW: 4.8980E+06 Hz
PM: 90.5224 deg
Av CM: -51.9763 dB, BW: 76643.3 Hz
CMRR: 88.061 dB

```

Figure 12: Results when using higher area

2.4 4. Common-Mode Rejection Ratio (CMRR)

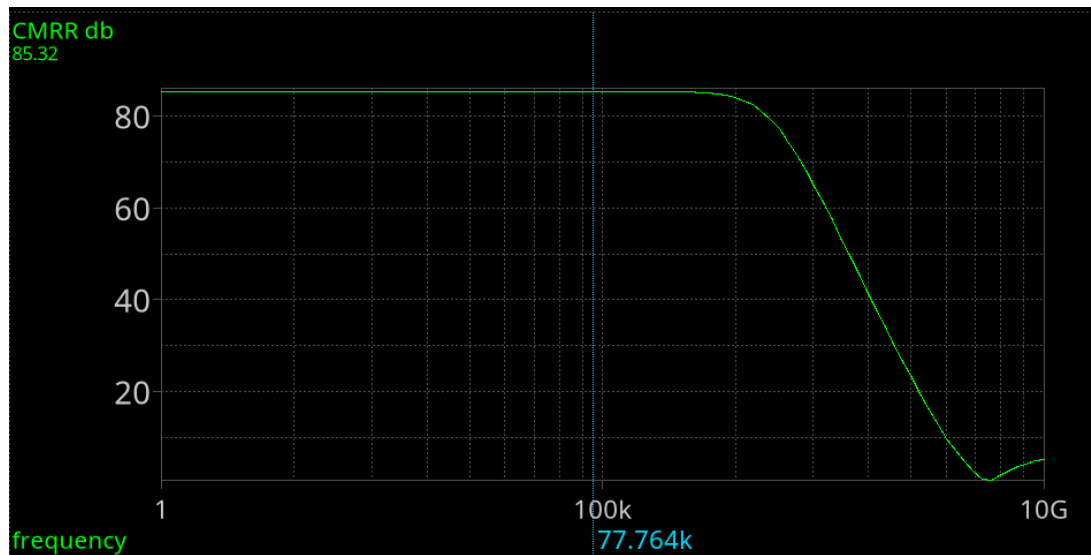


Figure 13: Common Mode Rejection Ratio

CMRR: Hand Calc vs Simulation

Parameter	Simulation	Hand Analysis
CMRR (dB)	85.321	83.151

2.5 5. A_{vcm} vs VICM

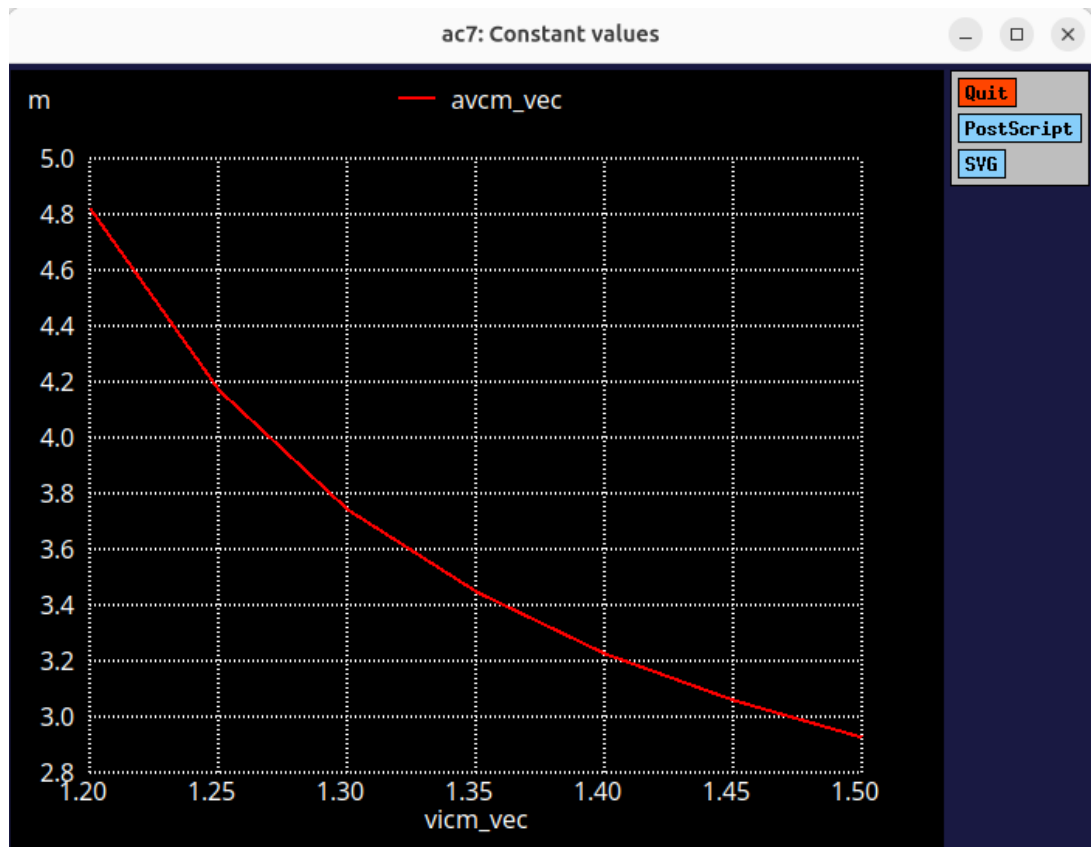


Figure 14: Small signal common mode gain versus common mode input

A_{vcm} Variation

Q: Justify the results. Does A_{vcm} increase or decrease vs. VICM? Why?

As VICM increases, V_{DS} of the tail current source increases, biasing it deeper into saturation, leading to increase in r_o and from the relation $\frac{1}{2 \cdot g_m \cdot r_o}$, A_{vcm} decreases.

2.6 6. Differential Large Signal Characteristics

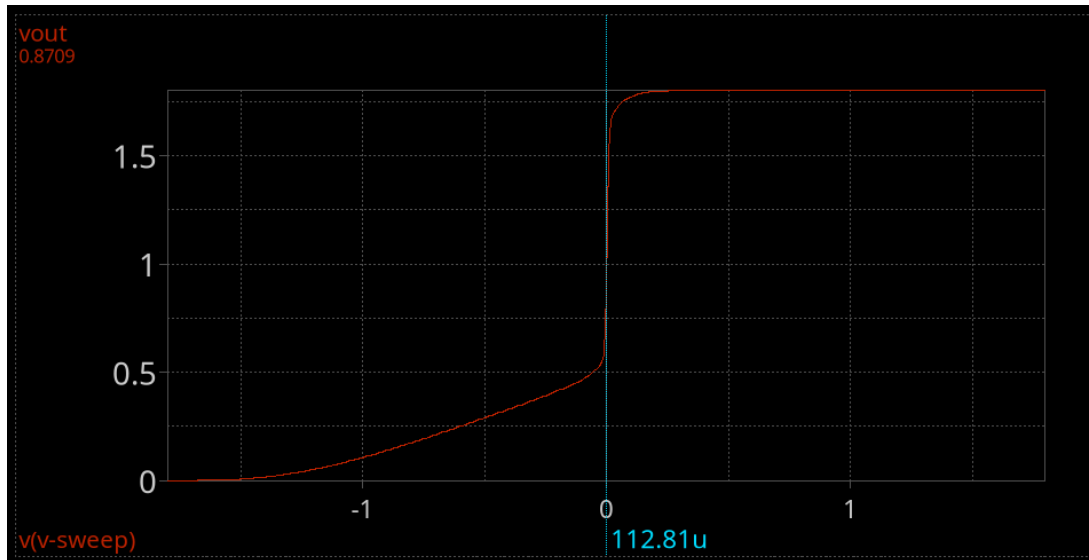


Figure 15: Vout vs vid

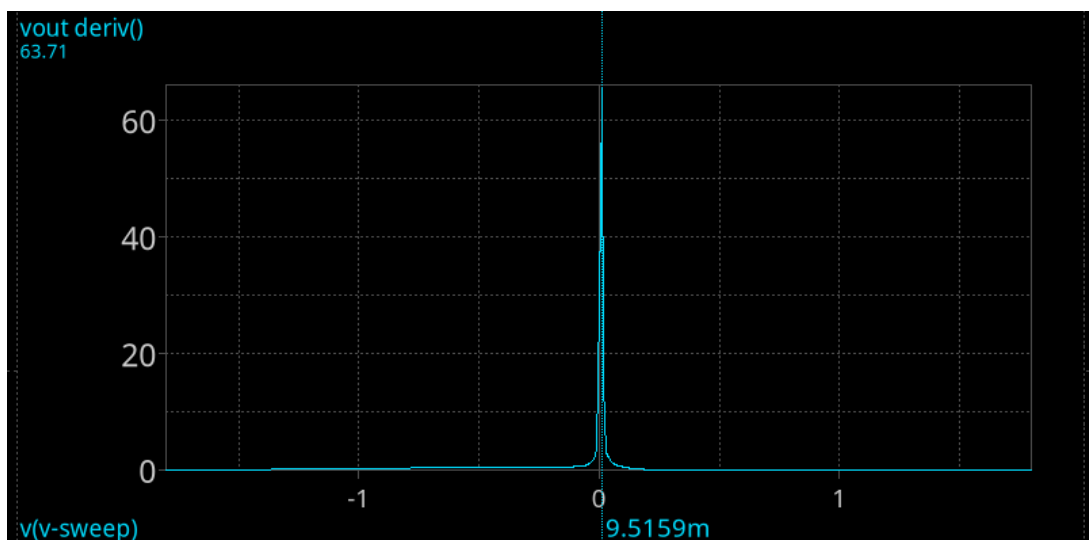


Figure 16: deriv(Vout) vs Vid

Differential Large Signal Analysis

Q: From the plot, what is the value of Vout at VID = 0? Why? Compare the peak of the derivative with A_{vd} . Comment.

At vid=0, $V_{out}=V_{OutQ}=0.8709V$. The peak of the derivative at 63.71 which is less than the actual diff gain from small signal analysis. The reason for this is that the step input differential is too large (10mV), which if used in higher gain OTAs, The peak will be much lower.

2.7 7. Common-Mode Large Signal Characteristics

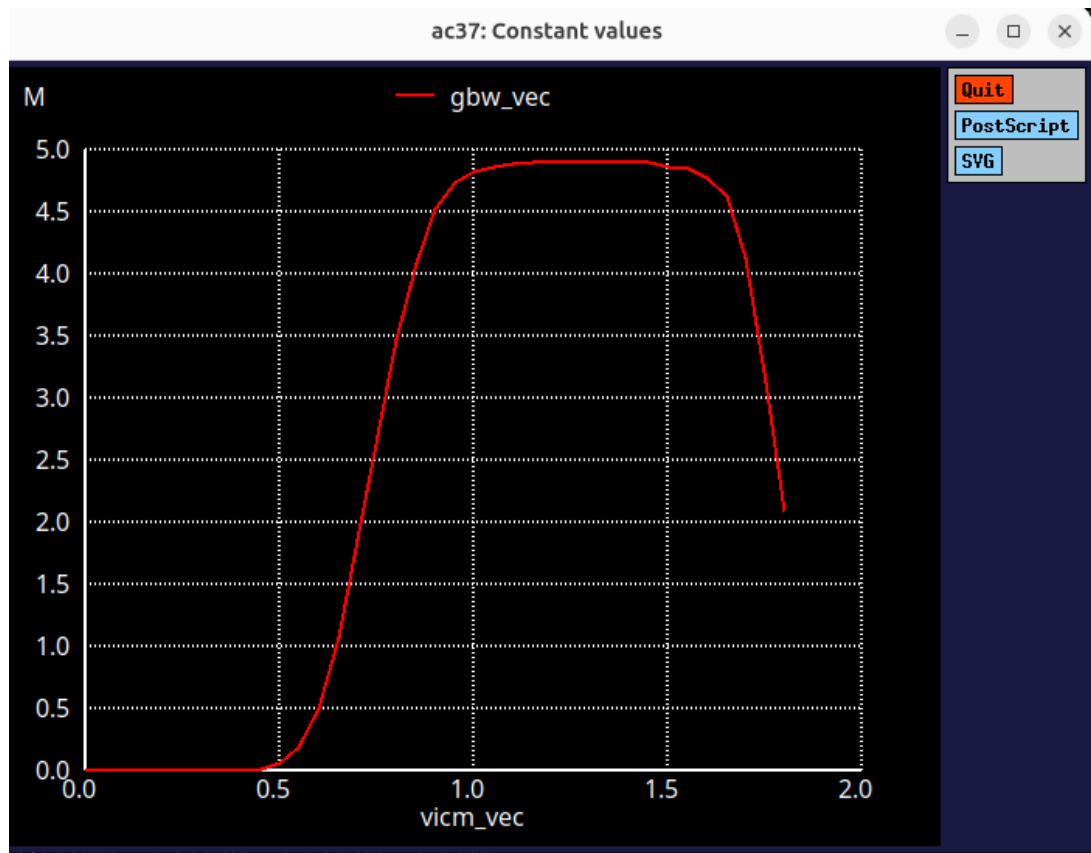


Figure 17: GBW vs large signal input common mode

CMIR Extraction

Q: Annotate the CM input range based on a 10% reduction in max GBW. What is the extracted CMIR? Does it meet the specs?

Extracted CMIR is from 0.95 V to 1.65 V, which meets the specs. The reason for CMIR extension is that when V_{iCM} decreases and V_{DS} of tail current source decreases, the body effect decreases which decreases V_{GS} , which limits the decrease in V_{DS} less than expected.

3. Part 3: Closed-Loop OTA Simulation

3.1 1. Unity Gain Buffer DC OP

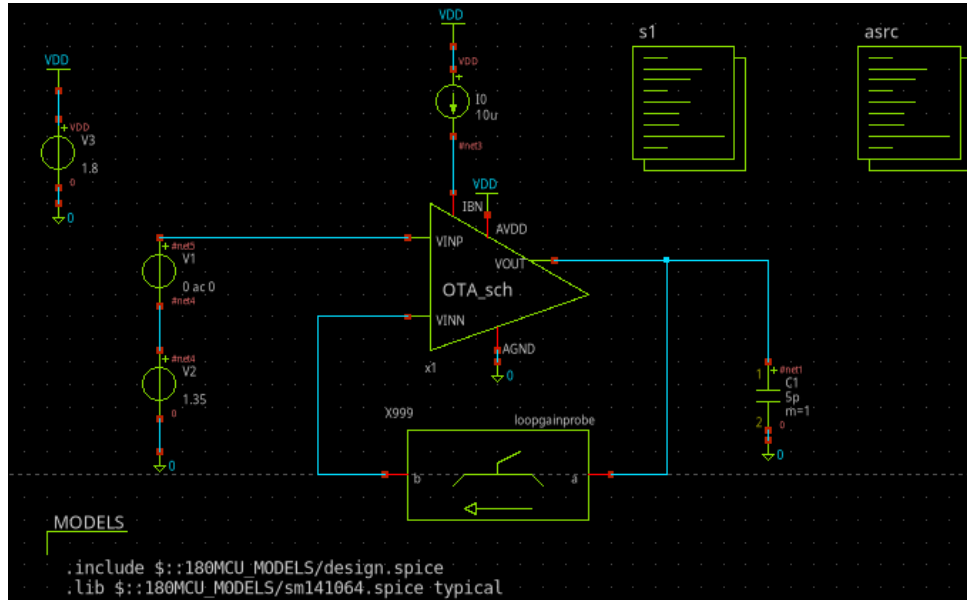


Figure 18: Closed Loop testbench

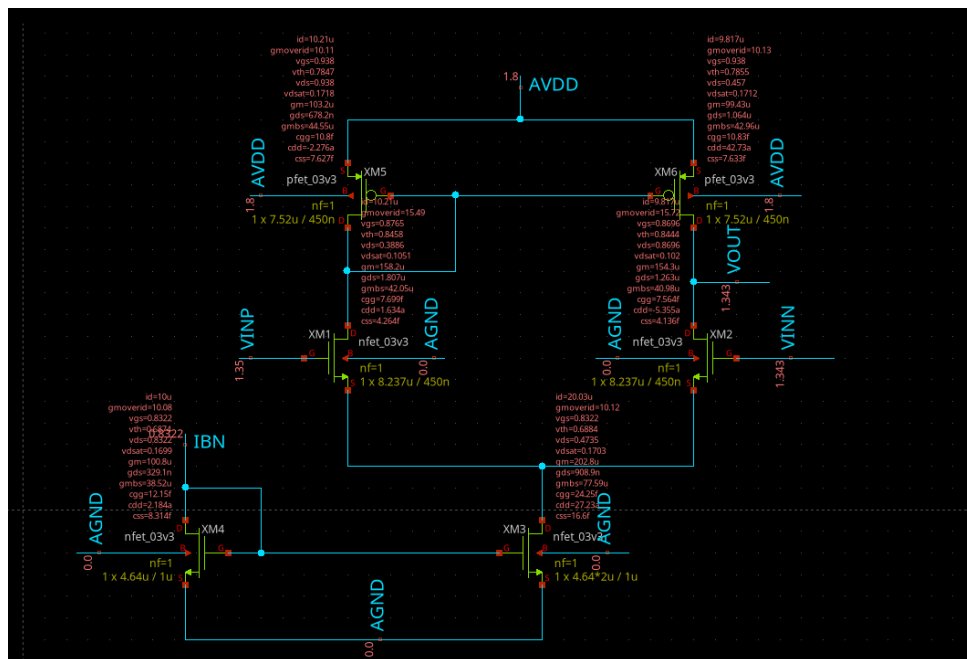


Figure 19: DC Operating points

Input Pair Mismatch Analysis

Q: Is the current (and g_m) in the input pair exactly equal? Why? Calculate the mismatch in I_D and g_m .

No they aren't the same because V_{out} is no longer equal to V_{mirror} so V_{DS1} is not equal to V_{DS2} , and V_{SD5} not equal to V_{SD6} leading to different currents and small signal parameters for both (systematic mismatch).

Note: In general Op-amps, It is crucial to simulate it in a closed loop to preserve V_{outQ} . Otherwise, if simulated in OpenLoop, any systematic or random offset will be amplified, leading to $V_{out} = +V_{DD}$ or zero. the 5T OTA is an exception in this as V_{out} is defined by $V_{mirror} = V_{DD} - V_{SG}$

3.2 2. Loop Gain (STB Analysis)

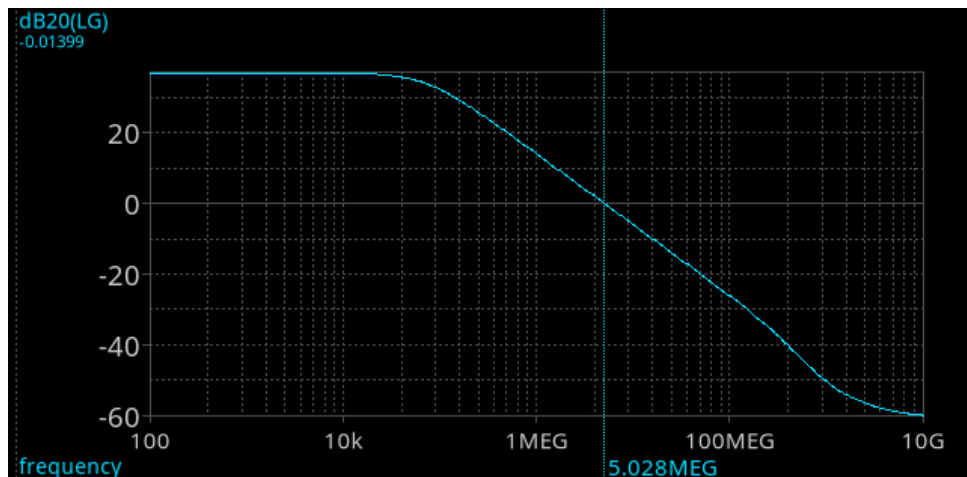


Figure 20: Loop Gain Magnitude in dB vs frequency

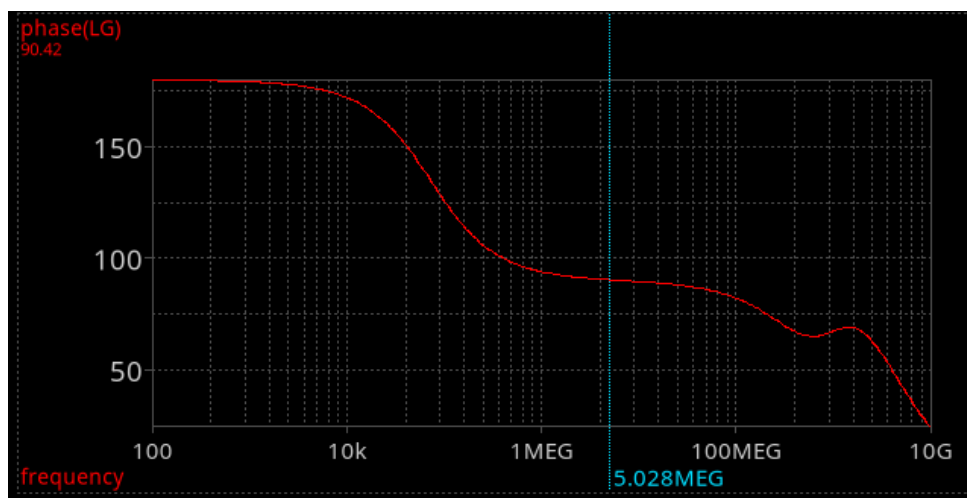


Figure 21: Loop Gain Phase vs frequency

Loop Gain Comparison

Q: Compare LG DC gain and GBW with those obtained from open-loop simulation. Comment.

Parameter	Hand Calculation	STB Simulation
DC Loop Gain	36.104	36.85
Gain CrossOver Frequency (MHz)	-	5.02
Phase Margin	-	90.418

4. Part 4: Effect of Mismatch on CMRR

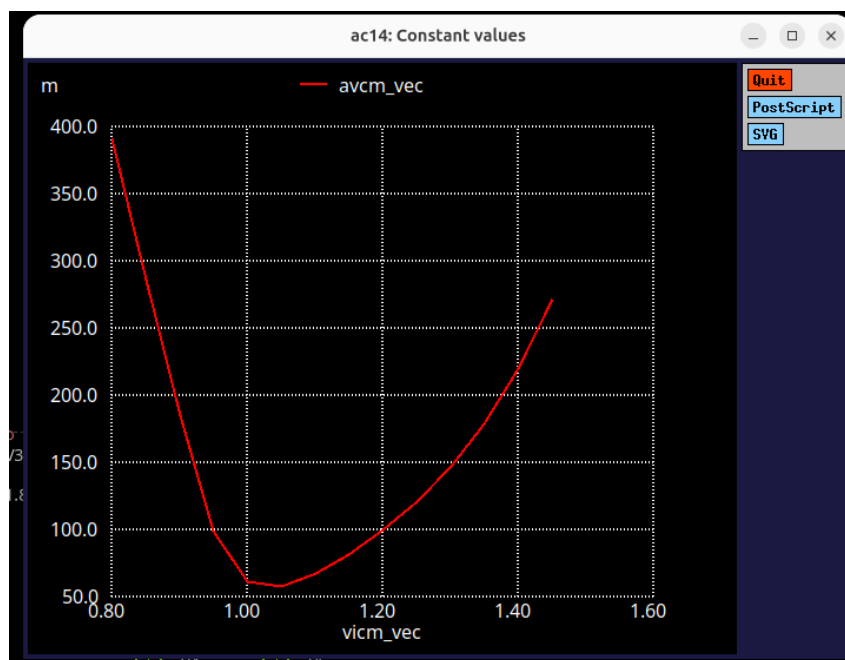


Figure 22: Closed Loop Avcm vs vicm

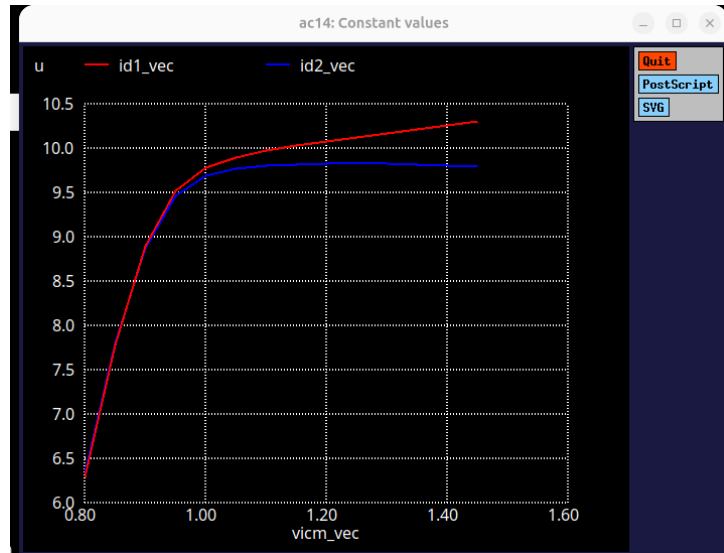


Figure 23: Current in XM1 and XM2 vs input common mode

Mismatch Effects

Q: Comment on the I_D of the input pair vs VICM results.

In Figure 22, when $\text{vicm}=0$, the tail current source is in triode, giving low R_{on} making the common-mode gain high. As the VICM increases, R_{on} increases, the CM gain decreases. Take care that ID1 and ID2 were the same. Then V_{out} will match V_{inn} . A mismatch occurs and ID1 not equal to ID2. ID2 will increase and ID1 will increase. The CM gain will increase as now the VF not equal to V_{out} resulting in δg_m

Parameter	Target Spec	Achieved Result
Open loop DC voltage gain	$\geq 34\text{dB}$	36.1 dB
CMRR @ DC	$\geq 74\text{dB}$	85.321
Phase margin (PM)	$\geq 70^\circ$	90.523
CM input range - low	$\leq 1.2\text{V}$	0.95 V
CM input range - high	$\geq 1.5\text{V}$	1.65 V
GBW	$\geq 5\text{MHz}$	4.96 MHz
OTA current consumption	Minimize	20 μA
Area	Minimize	28.1013 μm^2

Table 2: Design Specifications vs. Simulated Achievements